

Ryan F. Johnson, PhD
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United States Naval Research Laboratory
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Dear members of the Faculty Search Committee,

I am writing to apply for the [adapt for school]. I would be grateful if my application is considered for this job opportunity. My laboratory will specialize in computational physics to support future propulsion and energy conversion technologies. We will aim to develop predictive modeling capabilities for chemically reacting systems through high-performance computing, advanced algorithm development, and machine learning (ML)-driven methods. My research interests would be a valuable addition to the existing strength of interdisciplinary research in your department, including [adapt for school]. I also look forward to interactions with applied mathematicians, computer scientists, and physicists across the University.

I hold a B.S. and Ph.D. from the University of Virginia's Department of Mechanical and Aerospace Engineering and currently serve as a senior scientist at the U.S. Naval Research Laboratory (NRL), where I have been a civil servant since 2015. NRL operates as a working capital fund, requiring all research to be justified and supported through external agency funding or highly competitive internal seed programs. I am currently leading and funding teams of researchers through individually proposed and competitively awarded projects.

My research focuses primarily on using computational fluid dynamics (CFD) to explore propulsion systems and have pursued both fundamental scientific research and the translation of those insights to applied, system-relevant contexts. This past year, I was honored to receive the Presidential Early Career Award for Scientists and Engineers (PECASE), signed by President Joseph R. Biden, in recognition of these contributions. My ongoing research focus and outlook are as follows:

High-Speed Propulsion Modeling with CFD:

- Developing state-of-the-art CFD to understand next-generation engines (e.g., detonation engines, ramjets, and scramjets)
- Designing GPU-based solvers for both scale-resolved and reduced-order simulations.
- Analyzing flow physics in propulsion subsystems (e.g., detonations, fuel injectors, and supercritical flows)

Numerical and ML-Enhanced Frameworks:

- Transitioning high-order finite element methods for reacting flows on unstructured grids for complex flow geometries.
- Embedding accelerated chemistry solver strategies and Neural Networks (NN) into CFD frameworks.
- Investigating Tensor networks (TN) and Fourier neural operators (FNO), for model compression and operator learning

Future Research Focus:

- Discovering and exploiting novel subsystem strategies to create disruptive propulsion systems.
- Creating collaborative multidisciplinary research initiatives for next-generation sustainable propulsion systems.
- Designing ML- and tensor network-driven CFD frameworks tailored to future computing architectures.

My expertise aligns well with the department's curriculum. I previously taught independently Applied Partial Differential Equations for two semesters at UVA. I am prepared to teach core courses such as [adapt for school], as well as advanced topics like numerical methods, CFD, chemically reacting flows, and hypersonics. I also welcome teaching outside my core areas, as it fosters new ideas and broadens my research perspective.

My laboratory will advance my outlined research areas through robust software development, with a strong emphasis on mentorship and inclusivity. As the group expands, we will cultivate a collaborative environment that brings together individuals with diverse technical expertise and personal backgrounds. The laboratory will serve as a hub for fundamental research, open-access tools, and training in simulation approaches, energy conversion, and propulsion modeling.

I would be grateful to be considered for this position and please do not hesitate to reach out with further questions,

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Research Statement

Ryan F. Johnson, PhD

My vision is to establish a computational physics program with international visibility at [adjust to call] focused on advancing propulsion sciences and energy conversion systems. My research is motivated by the pursuit of faster flight for transport and space access. A promising strategy is the combined cycle propulsion, where a vehicle transitions from traditional engines such as turbofans or turbojets to high-speed devices like ramjets, scramjets, detonation engines, and potentially rockets (Figure 1). My research has focused primarily on the high-speed air-breathing propulsion in which chemical and fluid dynamic time scales converge and the resulting flows are highly complex. I utilize computational fluid dynamics (CFD) to address key challenges in high-speed propulsion by unraveling and predicting the underlying physics using state-of-the-art numerical approaches with machine learning (ML)-based compression strategies.

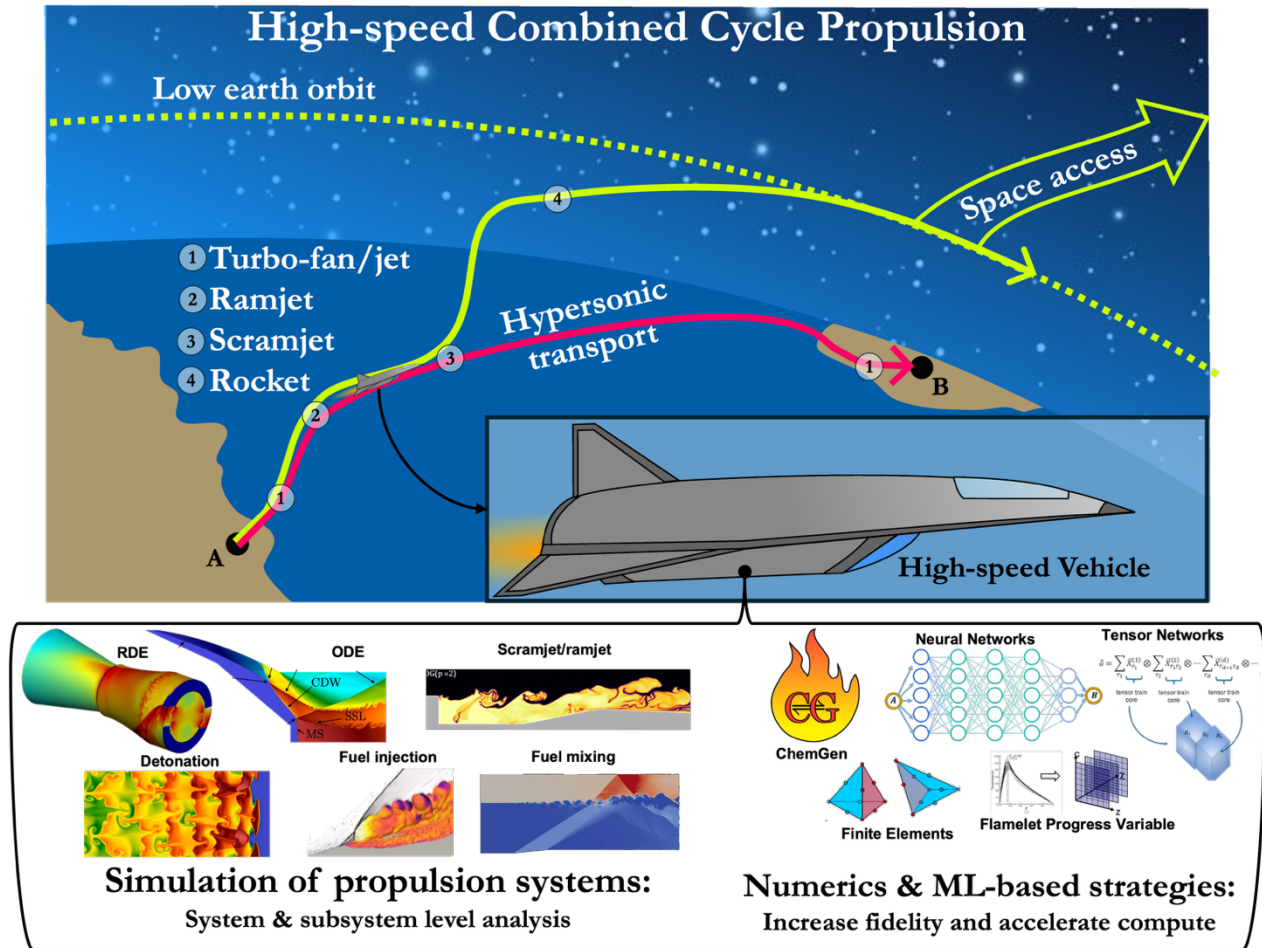


Figure 1. Current research focus of combined-cycle propulsion which uses high-speed airbreathers (e.g., ramjets, scramjets, RDEs, and ODEs) for rapid transport and space access. My research models these systems using state-of-the-art numerical and ML-based methods to accelerate accurate solutions.

Currently, I serve as a senior scientist at the U.S. Naval Research Laboratory (NRL), leading teams of researchers in a wide range of research topics. These are interdisciplinary teams; they contribute to both fundamental research and applied system design and development at the intersection of propulsion, chemically reacting flows, and advanced computational methods. Collectively, these experiences have prepared me to pursue a faculty position where I can expand my research interests and make a broader scientific impact. I plan to lead research across fundamental and applied domains all while mentoring the next generation of scientists and engineers.

My research interests and philosophy align strongly with [adjust to call]. I aim to contribute new computational capabilities to the department's strengths in propulsion and energy conversion, grounded in both fundamental and practical insight into system-level challenges and state-of-the-art modeling approaches. I anticipate collaboration with faculty across experimental (e.g., optical and laser diagnostics), theoretical (e.g., chemical kinetics, nonequilibrium thermochemistry), and computational (GPU acceleration, numerical integration, machine learning) domains. Beyond the department, my history of working across NRL's different focus area groups positions me well to engage researchers in computer science, applied mathematics, and related

engineering fields. I am especially motivated to grow these interdisciplinary partnerships, as some of my most rewarding research has been in interaction with individuals from various fields and academic backgrounds.

Research Background and Direction

My research focuses on high-fidelity, scale-resolving simulations of air-breathing propulsion systems, which operate by ingesting surrounding air, compressing it through an inlet, injecting and burning fuel in a combustor, and eventually expanding the exhaust through a nozzle which generates thrust, see Figure 2. Within this domain, I have performed research of three promising air-breathing systems: solid-fueled ramjets (SFRJs) [1], supersonic combustion ramjets (scramjets) [2], [3], and detonation engines (rotating detonation engines, RDEs [4], and oblique detonation engines, ODEs [5]). Of these systems, only lower-speed ramjets have been consistently fielded; scramjets and detonation engines are under active development for high-speed propulsion, with only limited flight demonstrations to date. Figure 2a-c shows large-scale simulation results SFRJ, scramjets, and RDE along with corresponding inlet[6] (Figure 2d) and nozzle[7] (Figure 2e) from my research.

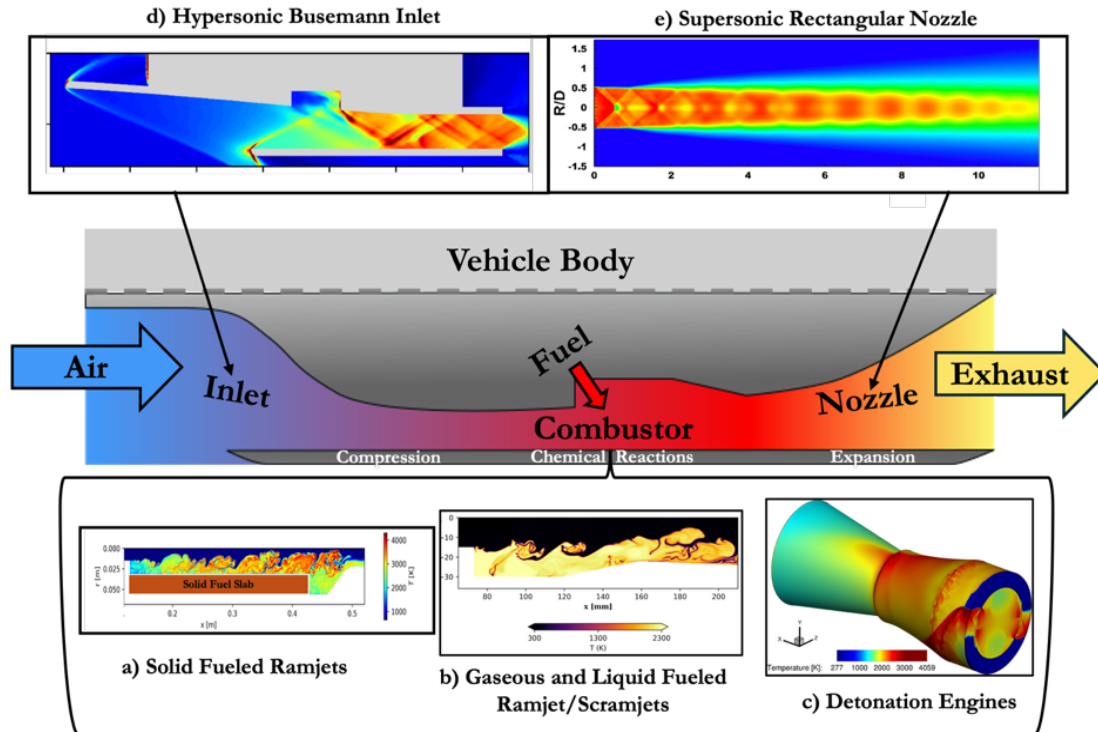


Figure 2. Air-breathing propulsion system schematic with accompanying system level CFD simulations: (a) SFRJs [1] (b) [2][3], (c) Rotating detonation engines (RDEs)[4] and oblique detonation engines (ODEs)[5], (d) Hypersonic Busemann inlet[6], and (e) Supersonic nozzle[7]

My research led to the development of chemically reacting flow simulation capabilities using finite element methods (FEM). Finite element methods are a promising approach for the simulation of propulsion devices due to their inherent ability to handle complex geometries using varied cell types and unstructured grids, simplifying meshing and reducing design time. The solver development I focused on was specifically optimized for high-performance computing on graphical processing units (GPUs)[8]. To my knowledge, my contribution was among the first to support chemically reacting flow simulations on memory-constrained GPU architectures. The tools I developed have been well recognized and transitioned to other laboratories. Despite advances in system-level simulations, uncertainties in model accuracy and high computational cost continue to limit the computational scalability, motivating disruptive advances in both modeling of the physics and developing of computational strategies.

Below, I outline two main research thrusts: subsystem analysis and numerical methods with ML-enabled strategies. Together, these efforts aim to support a broader, long-term goal of developing a physics-driven, software-development-based research program optimized for exascale computing architectures with focus on peak performance and model fidelity. The overarching goal is to develop a state-of-the-art multiphysics, multiscale framework to uncover design-relevant physics and deliver scalable, predictive tools for next-generation design capabilities for high-speed propulsion systems.

Research Thrust 1: Subsystem Analysis

In solid-fueled ramjets and scramjets, compression is achieved through inlet that generate shocks and raise pressure, followed by combustion in stabilized flame-holding regions. Detonation engines, in addition, use detonation waves to simultaneously perform compression and combustion in the mixture. These engines are driven by complex chemically reacting flow subsystems. My research focuses on the interplay of these processes and their impact on system-level behavior (Figure 3). These regions feature sharp gradients and fast reactions, and their growing complexity and numerical cost restrict the exploration and prediction of these complex physical phenomena, leaving their influence on overall design still in high uncertainty.

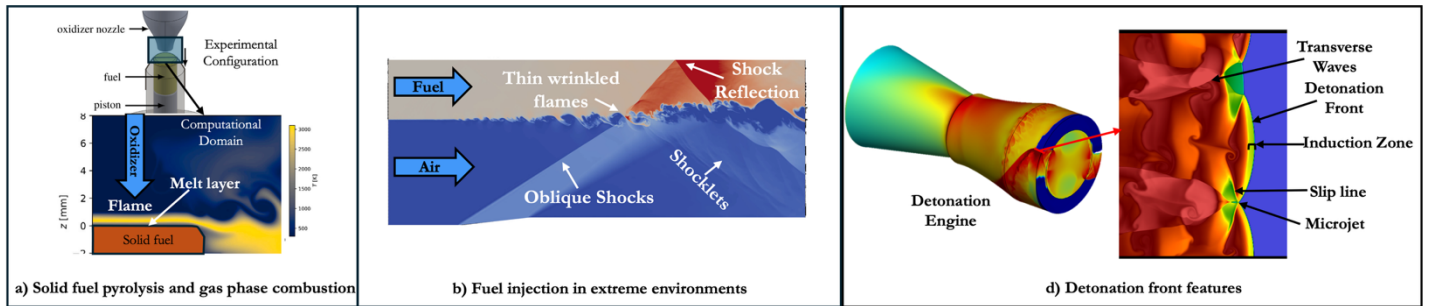


Figure 3. Complex physical phenomena found in propulsion systems: (a) Solid fuel combustion [9] [10], (b) Fuel injection in extreme environments [11] [12][13], and (c) Detonation front features [14] [15]

Solid fuel combustion: In the area of solid fuel combustion, my research has focused on CFD approaches that incorporate the complex physics, ranging from melt layers to fuel pyrolysis, found at solid fuel boundaries[9]. I have also investigated the fundamentals of gas-phase combustion coupled with the initial solid fuel pyrolysis process[10] (Figure 3a). Many aspects of solid fuel combustion remain unresolved due to its multiphase and multiscale nature, and the complexity of the underlying physics, making it a continued focus of my research.

Proposed Future Work: I propose research on more detailed solid fuel surface models, where pooling of the liquid melt layer can introduce complex physical effects on the flow field and near-wall flame propagation. This will enable a better understanding of how solid fuels can be integrated into propulsion systems and be accurately modeled in large-scale simulations.

Mixing of fuel under extreme conditions: I have performed scale-resolving simulations to better understand the relevant physics of mixing found when fuel is injected into high-speed environments (Figure 3b), with emphasis on new numerical methods to explore the interplay between flow compressibility with chemical reactions [11] [12]. In many cases, the injected fluid behaves as a non-ideal gas, and modeling its mixing remains beyond most available tools. I have built research programs to investigate the mixing behavior of these non-ideal behaving gases with focus on stability methods [13].

Proposed Future Work: I propose research projects for more complex fuels and fuel types, such as jet fuels with additives, and move towards more system relevant fuel injection systems, such as intersecting strut injectors. This will enable a system-level understanding of complex fuels and geometries which is crucial for next-generation engine design.

Detonations: My research has focused on implementing advanced techniques to accurately capture detonation physics using high-order finite element methods (FEMs). The flows involve shock waves coupled with intense exothermic reactions, forming rapidly traveling detonation front that gives rise to multidimensional features (Figure 3d). My research demonstrated the viability of FEM for detonation modeling and has more recently focused on enhancing method stability to ensure robustness and predictive confidence [14], [15], as discussed further below in Research Thrust 2.

Proposed Future Work: I propose research using an adaptive-fidelity, multi-method approach that integrates the advances in FEM with established finite volume methods (FVM). For detonation modeling, this balanced strategy is expected to better resolving the complex and poorly understood physics.

Research Thrust 2: Numerical Methods with ML-based strategies

To accelerate computational speed and to improve accuracy, my research will further focus on advancing numerical methods and ML-based compression strategies for propulsion problems. I have developed methods that mitigate numerical instability, maintain high-order accuracy, and reduce computational cost through ML-based compression techniques. These approaches

have enabled scalable, high-fidelity simulations and demonstrated previously unattainable results for relevant propulsion systems. Below, I outline details on these approaches and how I have actively transitioned them into chemically reacting flow simulations.

Structure preserving and finite element methods: In CFD, mixing chemical species in reacting flows can cause numerical instabilities. Traditional methods often violate conservation laws and sacrifice accuracy. My work demonstrated that finite element discontinuous Galerkin (DG) methods can be utilized in a fully conservative, high-order formulation that resolves instabilities in multispecies reacting flow simulations[14]. I have established collaborative research programs with computational mathematicians focused on structure-preserving methods (Figure 4a). These approaches offer promising advances for simulating chemically reacting flows by guaranteeing essential laws (e.g., conservative, positivity-preserving, and entropy-bounded) are upheld for the simulated solution. Our work has successfully transitioned these methods to unstructured finite element grids using the DG, Figure 4b, enabling demonstration cases not previously possible [11], [12], [13], [14], [15].

Proposed Future Work: I propose research to transition the new methods to complex physics, including plasmas and liquid-to-supercritical flows, enabling full FEM capabilities for next-gen engine geometries.

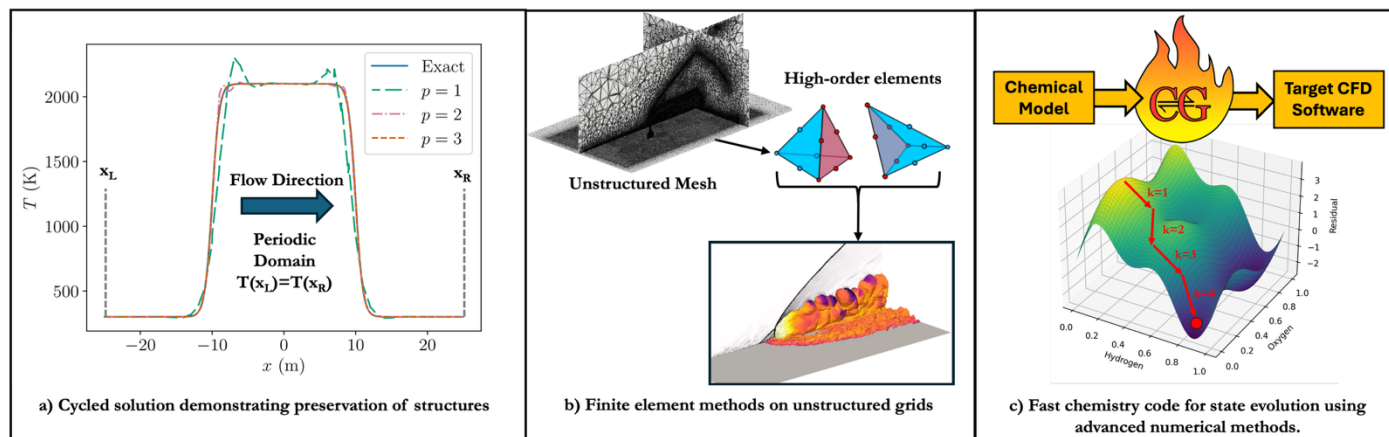


Figure 4. Numerical methods for complex physics: (a) Demonstration of structure preserving methods [12] [13] [14] [15], (b) Unstructured grids with high-order finite element methods [11], (c) Fast chemistry code using acceleration methods [16], [17]

Chemistry acceleration strategies: I have also focused on accelerating the integration of chemical kinetics into computational physics software. To this end, I developed ChemGen [16], [17], an open access flexible code generator that ingests chemical kinetic data and outputs tailored source code for specific computational ecosystems, Figure 4c. ChemGen enables customization of mathematical formulations and optimizations critical for simulating chemically reacting flows. A major barrier to combustion modeling is that kinetic mechanisms are often inaccessible to the broader CFD community. ChemGen addresses this by lowering the entry barrier for researchers transitioning from single-component CFD to reacting flow simulations. ChemGen provides various chemistry formulations and solver strategies. These strategies give the flexibility to improve computational efficiency and convergence.

Proposed Future Work: I plan to enhance ChemGen with compressed matrix-based solvers and transition them to large-scale propulsion simulations. These accelerated methods will remain open-access.

Progress variable approaches: High-fidelity models are often too costly for complex hydrocarbon fuels. To address this, I developed reduced-order and compression techniques to enable tractable system-level CFD simulations. My work includes advanced flamelet progress variable (FPV) methods, such as total energy FPV [18], and unsteady FPV [19], which project the full thermochemical state onto a low-dimensional manifold defined by a few transported scalars. These methods shift cost associated with large number of species and reactions to memory-intensive table lookups, with size scaling by fidelity. I have applied ML-based compression strategies to manage this growth, as outline below.

Proposed Future Work: I will extend progress variable methods to capture detailed physics, including pressure effects and plasma-based reactions, using compression strategies (see below) to maintain tractable problem size. This will enable large-scale simulations of devices with complex physics beyond simple chemistry, where detailed thermochemical states are intractable.

Embedded ML-based technologies: To enable ML integration into computational physics, I led the development of CodeJeNN [20], an open-access code generator that converts neural network (NN) data into source code for CFD and related applications. CodeJeNN has been applied from compressing FPV manifolds to fully replacing chemistry solvers in detonation simulations (Figure 5a). In parallel, I contributed to applying tensor networks (TNs) [19], to compress FPV manifolds, achieving

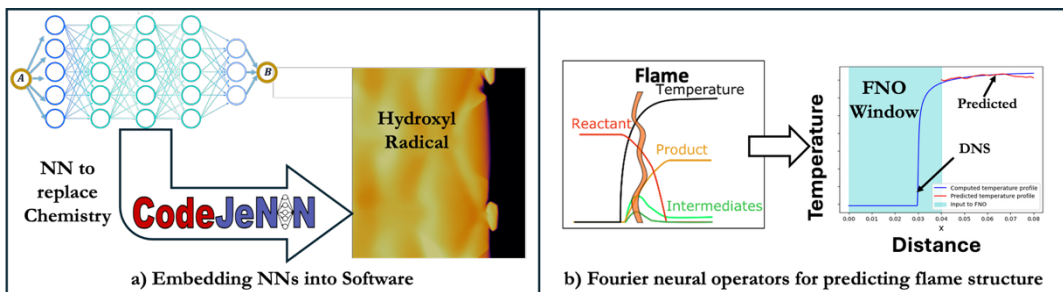


Figure 5. Compression methodologies for propulsion and computational physics

reductions from 800 GB to 100 MB, enabling simulations on memory-limited hardware. Unlike ML, TNs require no training and provides appealing error metrics. Recently, I also collaboratively explored Fourier Neural Operators (FNOs) [21], which learn operator mappings via Fourier transforms and can reconstruct flame structures with minimal input, potentially reducing mesh resolution needs (Figure 5b).

Proposed Future Work: I will continue to improve fidelity and accelerates computations by embedding NNs for acceleration and physical modeling purposes. I will also propose work to continue to pursue FNOs to predict flame structures in place of lookup tables, which could replace FPV entirely. Furthermore, I will use TNs to generate low-rank approximations of chemically reacting flows. These approaches can yield disruptive technologies that expand the range of physical problem sizes tractable for design.

Outlook

In my first five years, I will leverage existing connections and expand my network to pursue funding necessary for the research directions outlined above. I will build a research team of students and postdocs, and train them in state-of-the-art software development for propulsion and energy conversion, and mentor them in chemically reacting flow, CFD numerics, and ML-based compression strategies. Initially, we will extend existing open-source tools (e.g., OpenFOAM, PeleC, PyFEM, PyFR) and commercial platforms (e.g., CharLES, Converge, and Start-CCM) by developing the proposed features, while maintaining a focused strategy toward next-generation software development and predictive capabilities.

My laboratory will create new software that will implement a CFD-NN-tensor framework tailored for chemically reacting flows. Traditional CFD codes are not designed for NN inference, and current NN architectures remain underutilized in CFD algorithms. Bridging this gap will be disruptive, as future hardware will increasingly favor NN-centric architectures. By aligning with this trend, my proposed framework will deliver possible disruptive performance with a focus on delivering a better understanding of the complex physics unraveled by simulations that are previously impossible. This constitutes the second, broad objective of my laboratory. This NN-centric thrust will also enable on-the-fly learning during simulations, allowing integrated inlet-to-nozzle modeling and unlocking a new frontier in propulsion system design.

I will also broaden my research to include sustainable energy conversion strategies, an area I am also passionate about. I will develop new CFD models associated with sustainable aviation fuels (SAF). The chemically reacting flow technologies I developed have a natural extension to ground-level energy conversion, particularly in lower-speed regimes that I explored during my PhD work [22], [23] and with NRL collaborations [24]. In addition I have focused on distributed combustors for propulsion that already have many ground-use purposes for lowered emissions and efficient fuel usage [25]. Furthermore, my work on supercritical fuel injection would be relevant for sustainable energy applications, as SAF injection will likely involve non-ideal fluid behavior governed by the behavior of fluid at supercritical conditions (see work by Poblador-Ibanez and Nocivelli [26]).

Possible funding sources: I will leverage my connections at the Department of Defense, specifically targeting the Office of Naval Research, the Air Force Office of Science and Research, the Army Research Office for funding opportunities. I will invest in continued collaboration with my colleagues at Lawrence Livermore National Laboratory, where academic collaboration is encouraged through programs such as Academic Collaboration Team University Partnership. I will also pursue funding from the Department of Energy's ARPA-E and PSAAP programs. I look forward to focusing on fundamental scientific problems and proposing ideas to the National Science Foundation, i.e. the CAREER Program. Finally, I will also work to expand my network looking for industry partners, such as Intel, Nvidia, and AMD, as there is a growing interest in industry for demonstrating how emerging architectures can be utilized to accelerate computational physics. Over the years I have built relationships with faculty across the world, I intend to reach out to them as well to build projects together.

Final thoughts: Ultimately, my long-term goal is to develop multiphysics hybrid modeling environments that seamlessly combine high-order numerical solvers, reduced-order representations, and modern computing workflows. Such frameworks will unlock new frontiers for propulsion and sustainable energy conversion strategies, allowing for rapid iteration, uncertainty

quantification, and exploration of operating regimes that are currently inaccessible through traditional CFD or experimental means. I look forward to performing this research, all while growing a new workforce and contributing to humanity's ambition to fly faster, explore farther, and push the boundaries of human ingenuity.

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Education Philosophy

My education philosophy is supported by three core principles: passion, trust, and humility. I believe that these principles are essential to educator impact. Passion, I have found, is contagious. When an educator is genuinely excited about a subject, students are more willing to engage deeply, think critically, and push beyond their given material.

Trust is something that is reciprocal. I trust my students and mentees to take ownership of their education. That trust is reflected in my student-centered style built on collaboration. In return, students trust that I'm invested in their success. This creates a foundation of healthy learning as well as an environment suited for the patience required for complex tasks.

Finally, humility creates accessibility to the educator. It encourages questions and a comfortable dialogue that can quickly build a strong educational connection. My students and mentees knew I was always there to support them and that, despite my experience, I was open about my own imperfections and willing to learn alongside them. Having once been in their position, I could empathize with where students struggled and worked to address those challenges directly.

These principles have shaped my teaching and mentoring. The experiences of teaching, some of which I outline below, have been some of the most gratifying experiences of my life. Those experiences are why I'm eager to return to academia: to build new, meaningful relationships while helping educate and continuing to mentor the next generation of engineers and scientists.

Teaching Experience

My teaching began with tutoring in high school and continued through my graduate studies at UVA, where I became a well-known tutor for advanced topics like fluid dynamics and thermodynamics. In 2013, I was awarded the selective Farragh Fellowship to co-teach the undergraduate Partial Differential Equations (PDEs) course with Professor Patrick Hopkins. At the time, the course had poor evaluations, 2.75/5.0, due to dense content and limited context. We restructured it with a student-centered approach based on my tutoring experience, front-loading key math concepts and clearly outlining our strategy. Our strategy addressed knowledge gaps from varied student backgrounds by reviewing prerequisite mathematics and offering additional office hours to ensure all students were comfortable with the foundational material. I added a computational module using MATLAB and Python, enabling students, many with no computational background, to solve physically meaningful PDEs by the end of the semester.

Beyond restructuring, Professor Hopkins and I implemented a flipped classroom approach. For part of each session, students worked through introductory-level homework problems in small groups, creating collaborative discussion before we reconvened to address questions and clarify misunderstandings. By the midpoint of the course, I was delivering full lectures to a class of nearly 45 students. I incorporated real-world analogies, relating wave equations to guitar strings, the method of characteristics to shock waves, the heat equation to a hot pan, and spherical coordinates to droplets.

Our efforts led to a complete shift in student engagement with PDEs, reflected in a 4.7/5.0 average course evaluation. Students consistently offered positive feedback, including:

“Very good class, I feel like I have a good mastery of the material as a result of excellent organization and explanation by the instructors.”

“They took a subject [initially] I thought I would hate, and made it something I love. I can't thank them enough for being the amazing instructors that they are.”

The following year, the chair of the UVA Applied Mathematics Department invited me to teach the PDE course with full autonomy, a rare opportunity at UVA, where courses are typically taught by full-time faculty. I retained the student-centered approach from the previous year with a few adjustments. The major change was addressing the lack of an accessible undergraduate PDEs textbook. After teaching in Fall 2013, I wrote my own textbook: Handbook of Partial Differential Equations, a 175-page text with eight chapters, figures, homework problems, and example code for computational methods. The book remains freely available for collaboration [here](#). The course's popularity grew from 2013 and led to nearly 70 students for this 2014 semester. It was a demanding term, but I tried to interact with each student and expanded office hours to meet the

increased demand. This semester was another success, as reflected by the increased rating to 4.88/5.00. Almost all the reviews were positive including feedback for my book:

“This class was extremely valuable, and I really like the current layout of the class. The handbook written by Dr. Johnson was so, so helpful in the understanding of the material”

And some feedback demonstrating my teaching philosophy resonated well:

“The fact that he was so relatable (not to mention knowledgeable) made students want to come to his class, to ask questions, to be curious. So definitely, for the next instructor, listen to what Ryan says about this class, grab his materials and follow in his footsteps. He made learning fun again.”

My contribution as an instructor was commended by UVA when I received a Graduate Teaching Award that was nominated by the students and UVA’s professors. It was bittersweet when my time came to an end at UVA as an instructor to move to the Naval Research Laboratory (NRL). I look back fondly on those two semesters of teaching and am excited to return to lecturing and fruitful experiences that come with it.

Mentorship Experience

Since joining NRL in 2015, I’ve mentored over twenty students across high school, undergraduate, and PhD levels. I tailor each internship to match the student’s background: early-stage students gain exposure to applied physics and develop coding and analysis tools, while advanced students produce technical deliverables, some resulting in NRL memoranda or peer-reviewed papers. Many have gone on to graduate programs at top institutions with highly competitive graduate fellowships or positions in national labs and tech industry.

As my role expanded to managing multiple research efforts, I began mentoring new hires, guiding them through early career challenges and helping them focus on meaningful contributions. Several now lead independent projects, give invited talks, and help shape the direction of our group. My impact at this level was recognized by NRL leadership and contributed to my recent promotion to senior scientist. In addition to my scientific contributions, mentorship was a key factor in the promotion decision, making me the fastest employee to achieve the senior scientist rank in my division, within less than 10 years of service.

During my 2024-2025 sabbatical in Stanford’s Mechanical Engineering Department, I mentored graduate students in the labs of Professors Hai Wang and Beverley McKeon. I assisted both labs by teaching the theory of computational fluid dynamics and its interaction with combustion phenomena. Given the alignment with their research, my expertise provided valuable support to their ongoing projects. I met with some of their students frequently to guide research direction and troubleshoot complex simulations. These collaborations have produced several papers now under review, in diverse subjects ranging from chemical kinetics to machine learning. My connection with these students and their advisors has continued since my return from sabbatical, and I still advise them to assist in making positive impacts on their research.

Outlook

I look forward to returning to teaching and contributing to an engineering department through a range of undergraduate courses, including fluid mechanics, aerodynamics, thermodynamics, propulsion, computational methods, version control, linear algebra, ordinary and partial differential equations, and calculus (introductory to advanced). At the graduate level, I am prepared to teach courses in computational fluid dynamics, numerical methods, combustion, gas dynamics, chemically reacting flows, advanced fluid mechanics, hypersonics, and high-performance computing. I aim to structure courses to be student-driven, continually adapting to how students learn most effectively and bring in as many real-world analogies from my own work experience. ____ **list course codes.**

When building my research group, I will adopt a collaborative environment where students grow into confident, capable scientists and engineers. I look forward to leading a group composed of individuals from diverse backgrounds and perspectives, where constructive discourse arises naturally as we work together to solve challenging problems. In addition to developing their expertise in computational physics and propulsion, I will help them become stronger presenters, writers, and collaborators. As my colleague Professor Hai Wang once said to me, after I called him brilliant, “I am not brilliant my students are brilliant.” I hope to follow that attitude by mentoring a generation whose collective contributions exceed my own.